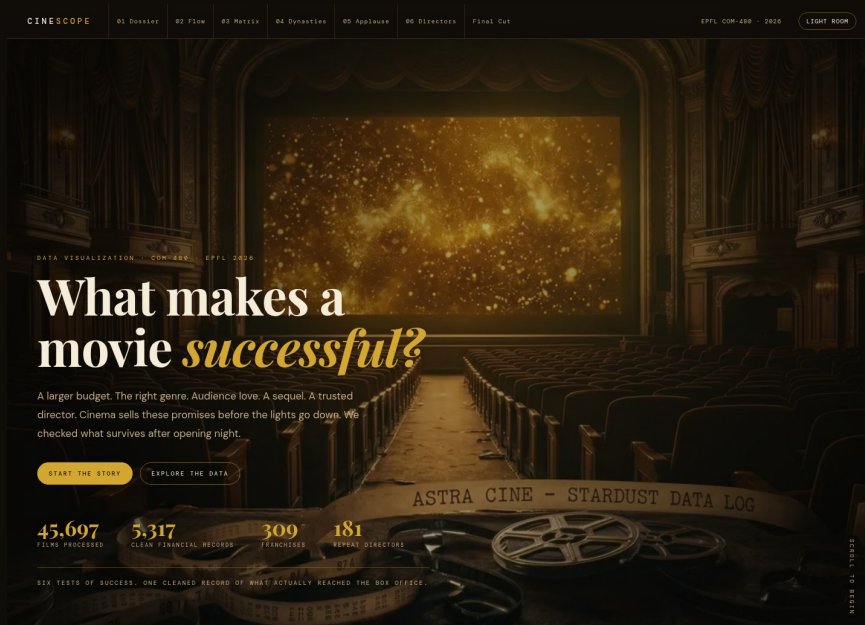


What makes a movie *successful*?

A movie can buy scale, earn applause, or inherit a name.
None of them buys certainty.



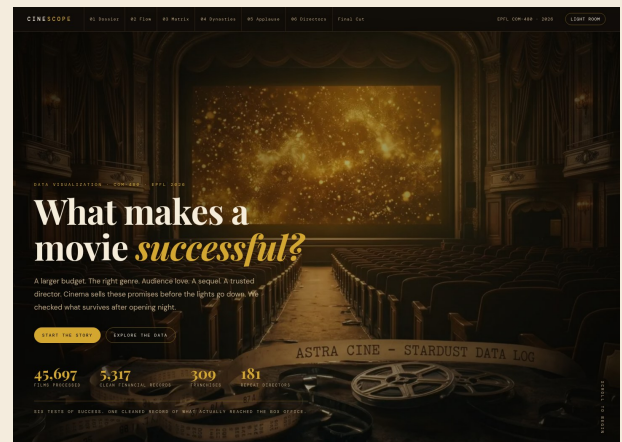
From an industry promise to an evidence trail.

Films arrive surrounded by persuasive explanations for success: spend enough, choose the right genre, win audiences, continue a franchise, attach a known director. Our original question was broad: **what drives financial and critical success in cinema, and how has it shifted over time?**

For the final story we turned that question into six tests that audiences can follow in sequence. The reader starts with individual films, aggregates them into production routes, tests inherited and audience-driven advantages, and ends at directing track records. The audience is any cinema viewer who recognises these promises but wants to inspect the receipts rather than accept mythology.

Narrative arc

Scale → route → genre fit → inheritance → applause → reputation.
Each step asks whether a plausible advantage removes uncertainty.



Final opening: a dark screening room frames the six “promises” before any chart appears.

THE SIX SCENES

- **Dossier Board:** does spending predict receipts?
- **Flow:** which routes repeatedly reach outcomes?
- **Matrix:** does scale suit each genre equally?
- **Dynasties:** do sequels inherit success?
- **Applause:** is admiration commercial power?
- **Directors:** is reputation repeatable?

45,697

PROCESSED FILMS

5,317

FINANCIAL RECORDS

309

FRANCHISES

181

DIRECTORS WITH 6+ FILMS

A threshold before a conclusion.

The Movies Dataset joins TMDb metadata, credits and MovieLens ratings. Financial fields are sparse and some recorded budgets are implausibly tiny, making ROI explode for clerical rather than cinematic reasons. We therefore documented one defensible rule: financial views require positive revenue and a recorded budget of at least **\$10,000**. This removes 65 near-zero entries while retaining *Paranormal Activity* at its reported \$15,000 budget.

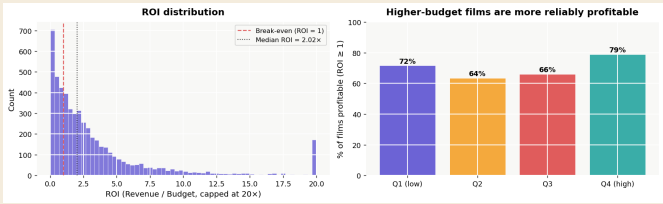
Universe	5,317 unique films with usable financial evidence
ROI	worldwide gross / recorded production budget
Ratings	MovieLens comparison requires 50+ ratings
Franchises	each eligible sequel compared with the collection original
Coverage	financial releases through 2017

Of 45,697 processed records, 5,417 contain positive budget and revenue. After the floor and 35 duplicate eligible IDs are removed, **5,317 unique films** support all financial claims. A MovieLens minimum of 50 votes leaves 4,607 films for comparisons involving audience response.

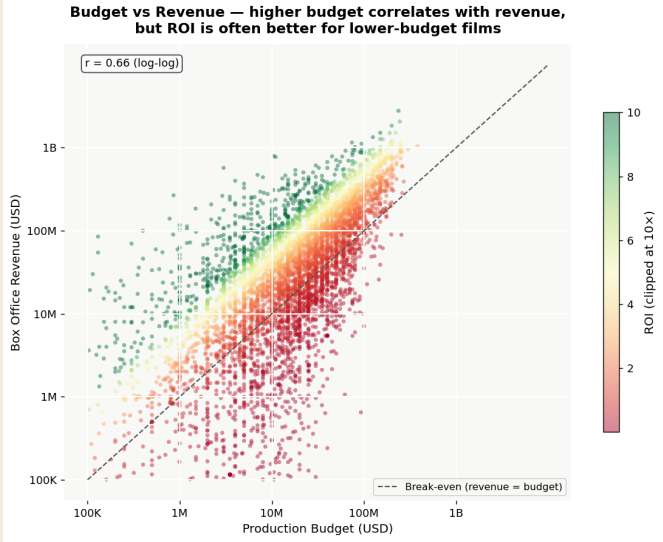


WHAT THE MEASURES CAN CLAIM

Gross divided by reported production budget is an interpretable return proxy, not studio profit: marketing, distribution splits and inflation are unavailable. The site therefore presents observed associations and comparisons, never a causal recipe or a prediction.



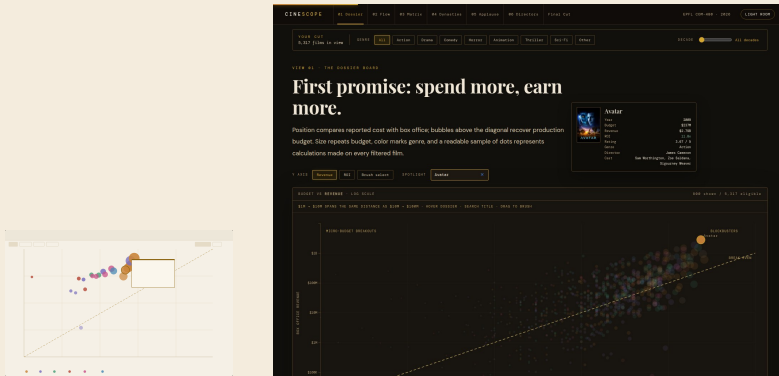
EDA identified the long ROI tail and the need for a stated lower-budget rule.



Budget and revenue call for log scales: their spread spans orders of magnitude.

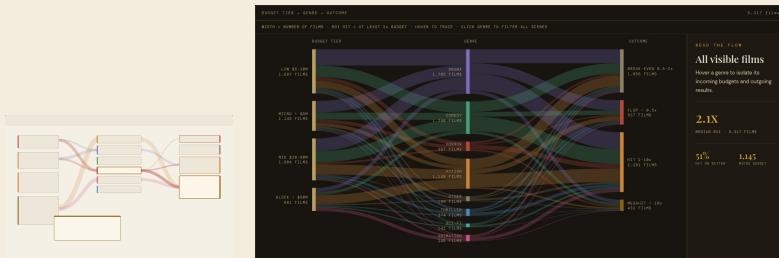
Scale becomes a route.

VIEW 01 / THE DOSSIER BOARD



M2 diagram → final searchable scatter and film dossier.

VIEW 02 / THE FLOW



M2 route proposal → computed Sankey with coordinated selection.

Reading it. Both axes are logarithmic: equal distance means a multiplication, not an equal dollar increase. The dashed diagonal is gross equal to budget; above it, gross exceeds the recorded cost. Toggle ROI, search a title, brush a cluster, or hover a mark for evidence.

Decision. Calculations retain all 5,317 films; only the mark layer is a deterministic 800-film sample so dense overlap does not erase patterns.

Reading it. Ribbon width is film count, flowing from budget tier through principal genre to ROI outcome. Hovering isolates routes; clicking a genre carries that investigation through the other views. The revenue/ROI switch separates reach from efficiency.

Interpretation. Expensive production widens access to large grosses, but micro films earn a higher typical multiple: **2.98x** median ROI versus **2.24x** for blockbusters.

Story turn

A large budget is a route to scale, not a guarantee of return. The next view asks whether the same budget behaves differently when genre changes.

EXPLORATION PATH

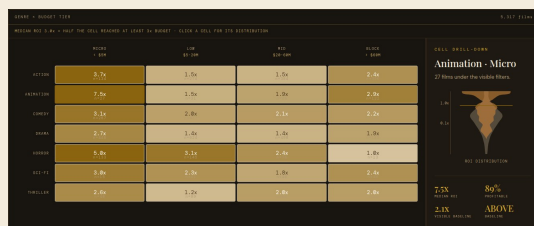
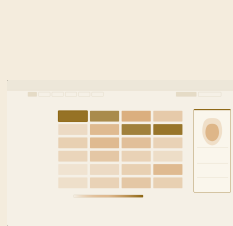
A reader can hover an outlier in the Dossier Board, toggle from gross to ROI, then click its genre in the Flow. The question consequently shifts from “is this title exceptional?” to “is this production route repeatedly favourable?” This was the reason for coordinated controls rather than four disconnected charts.

WHAT WE RESISTED

A poster mosaic or thousands of full-opacity bubbles would make a lively screen but a poor comparison. We kept images in requested detail only, faded guides, separated overlapping marks with thin strokes, and allowed aggregates to represent the complete eligible sample.

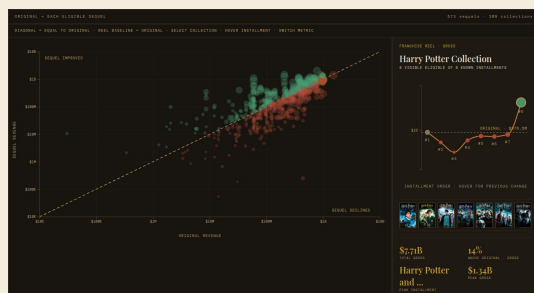
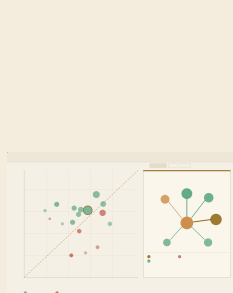
Fit, then inheritance.

VIEW 03 / THE PROFITABILITY MATRIX



Proposed grid → measured heatmap with distribution drill-down.

VIEW 04 / DYNASTIES



First-sequel sketch → every-sequel scatter and chronological reel.

Reading it. Rows are genres and columns are budget tiers. Colour is either median ROI or the share whose gross recovers budget. Click a cell to expose count and spread; change decade to test whether a pattern survives time.

Interpretation. Horror leads displayed genres overall at 3.01x median ROI and 77.7% recovery. Within micro budgets, its median reaches 4.95x over 133 films: a genre pattern, not one exceptional title.

Reading it. Each point compares an installment to its original; points above the equal line grew. Selecting a collection opens every eligible installment in order, with gross, ROI and rating modes.

Redesign and result. Limiting the sketch to sequel one would miss later recovery or decline. Across 573 sequel comparisons in 309 collections, only 46.6% outgross the original. A franchise is access to an audience, not inherited certainty.

Story turn

Genre can alter the odds and an original can create demand. Neither answers whether admiration or a famous name reliably converts into receipts.

FROM SKETCH TO MEASUREMENT

The M2 heatmap contained hypothesised numbers to establish layout. The finished cells are calculated from filtered JSON and expose their n , so a high-return cell with few observations cannot quite carry the argument. The decade filter tests persistence rather than freezing one historical market.

WHY ALL SEQUELS MATTER

First sequels measure immediate continuation only. Collections frequently peak later, restart or fade after several entries. Plotting every eligible sequel against the same original creates one comparable baseline while the reel restores order; on-demand installment posters identify an opened collection but never replace the values.

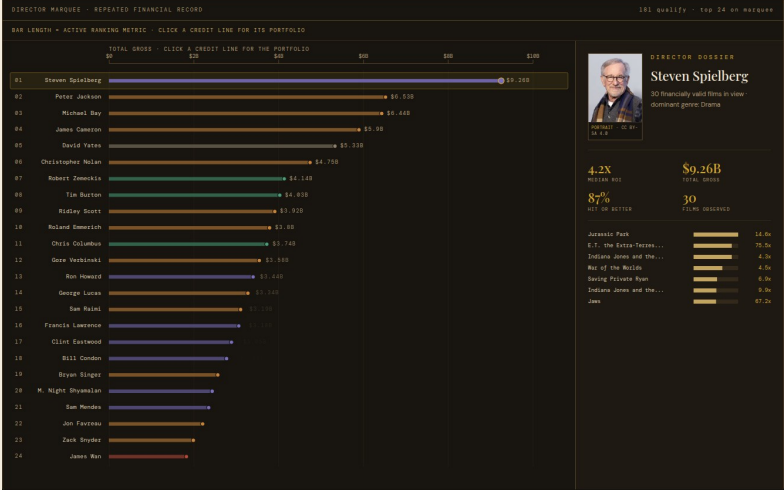
Approval and reputation do not close the case.

VIEW 05 / APPLAUSE VERSUS RECEIPTS



Bars encode the middle 50%; diamond markers encode median gross; points retain films.

VIEW 06 / THE NAME ABOVE THE TITLE



A ranked marquee plus selectable portfolio dossier; threshold is adjustable.

Why it was added. Financial success and audience response are parallel outcomes in our original question, but the M2 sketches mainly tested money. This view makes that missing comparison explicit.

Reading it. MovieLens score bands run vertically; horizontal position is gross on a log scale (or ROI after toggling). Hovering a dot retrieves the film behind a band.

Result. Across 4,607 films with 50+ ratings, rating and logged gross are almost unrelated: $r = 0.017$. High applause is meaningful success, but not a receipt forecast.

Reading it. Rank directors with at least six eligible films by median ROI or total gross, then select a row to inspect the underlying titles. Requiring repeated work prevents one breakout from being treated as a career.

Result. James Wan leads median ROI at 8.00x over seven films; Steven Spielberg leads accumulated gross at \$9.26B across thirty. Even reputation changes meaning when the success measure changes.

Why two new views?

The four M2 views explain production strategy and legacy. These additions close the promised critical-success question and turn “bankable name” into a measurable, inspectable claim.

CARE WITH APPLAUSE

Ratings are not reviews from every ticket buyer, and older or niche films can attract different MovieLens audiences. Requiring 50 votes removes the least stable averages; it does not convert association into causality. The overlap of distributions is therefore more honest than a single correlation alone.

CARE WITH REPUTATION

Director credits coexist with franchises, studios, casts and release contexts. A minimum-film control makes consistency inspectable, while the two rankings prevent “bankable” from silently meaning either efficient budget use or sheer accumulated box office, depending on which result looks strongest.

A theatre interface, a modular engine.

TECHNICAL STRUCTURE

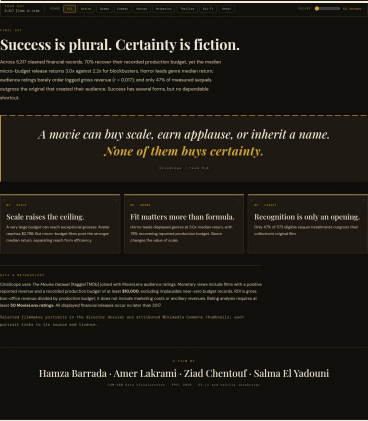
- Semantic single-page narrative in `index.html`; dark-room CSS system with a light-room option.
- D3 v7 modules under `js/viz/`: scatter, Sankey, heatmap/distribution, sequel scatter/reel, interval plot and ranked marquee.
- Shared genre and decade state coordinates the scenes; local controls change each scene’s metric.
- A reproducible Python export produces browser JSON after the documented budget threshold and franchise construction.
- Contextual imagery stays subordinate to data: current TMDb posters are available on demand for named dossier evidence and installments in the 100 highest-grossing collections; attributed Commons portraits appear in the director dossier.

```
data/processed/movies.csv
↓ export_web_data.py
movies.json / franchises.json
directors.json / summary.json
↓ shared application state
six independent D3 view modules
```

Coordination. A global genre and decade state updates the analytical sequence without hiding scene-specific questions. Revenue/ROI/rating switches remain local because they change the meaning of a view, rather than the set of films being followed.

CHALLENGES AND DECISIONS

Challenge	Response
Unstable ROI outliers	Documented \$10k budget floor; retained plausible breakouts.
Dense scatter	All films inform results; deterministic 800-film mark sample remains readable.
Sequel simplification	Replaced first-sequel idea with every eligible installment and a chronological reel.
Two meanings of success	Added revenue/ROI toggles and the audience-rating interval view.
Visual identity	Dark projection-room language, restrained animation and progressive detail panels.
Poster availability	Refreshed TMDb reel images cover the 100 largest collections and load only when opened; missing artwork creates no blank card.



The conclusion returns exploration to one memorable claim without claiming causality. **Validation.** The export script regenerates every JSON view

from processed data; local browser checks cover view rendering, filters, media failures and interaction paths. The generated process book is kept alongside its $\text{\LaTeX}$ source for reproducibility.

FINAL IMPLEMENTATION CHECKPOINT

The narrative is not a sequence of static claims: shared genre and decade controls change the views together; local metric toggles test alternate definitions of success; hover and selected panels return aggregates to individual film evidence. The final interface preserves the warm parchment language of the M2 sketches inside a dark projection-room setting designed for screen presentation and the screencast.

Several paths to success. No dependable shortcut.

Scale

Avatar reaches the largest observed gross, yet smaller-budget films post a stronger typical return than expensive productions.

Fit

Horror produces the strongest overall median ROI among displayed genres; budget only makes sense in relation to genre.

Legacy

A franchise name opens the door, but fewer than half of measured sequels outgross the original.

Approval

Ratings and gross answer different questions; applause barely orders revenue in this sample.

PEER ASSESSMENT / CONTRIBUTION BREAKDOWN

The final system was integrated collaboratively; primary responsibilities are recorded below.

Member	Primary contributions
Hamza Barrada	Data preparation and ROI methodology; coordinated site integration, final testing and repository documentation.
Amer Lakrami	Dossier Board and Flow interaction work; filter usability review and screencast preparation.
Ziad Chentouf	Profitability Matrix and Dynasties analysis; sequel comparison interpretation and visual validation.
Salma El Yadouni	Applause and Directors narrative views; editorial review, process-book and final presentation support.

SOURCES AND ACKNOWLEDGEMENT

Data: *The Movies Dataset* (Kaggle/TMDb) joined with MovieLens ratings. Film poster context is supplied by TMDb; this product uses the TMDb API but is not endorsed or certified by TMDb. Director portraits are attributed Wikimedia Commons assets with per-file licence links stored in the repository.

A movie can buy scale,
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None of them buys
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CINE SCOPE / TEAM XLB